

QUADRUPED ROBOTIC DOG

Engineering Design and Specification Report

Design Class: 4-Legged Robot (12–14 DOF)

Dimensions: 50–70 cm (L) x 30–40 cm (H)

Mass Budget: < 14 kg

1. Introduction

This report was prepared to document the mechanical, electrical, actuation, sensing, and control architecture for a quadruped robotic dog platform. The design targets an overall length of 50–70 cm, a standing height of 30–40 cm, a total of 12 to 14 degrees of freedom (DOF), and a mass budget below 14 kg. The design decisions documented below were derived from established quadruped robotics literature and open-source builds, with the corresponding reference cited at each design decision ^{[1][2][3]}.

2. Design Requirements and Specifications

Parameter	Target Value	Design Driver
Overall length	50 – 70 cm	Sets link length scaling and torso size
Standing height	30 – 40 cm	Sets leg link length and joint torque requirement
Degrees of freedom	12 – 14 DOF	3 DOF per leg (hip yaw, hip pitch, knee) + optional spine/head DOF
Total mass	< 14 kg	Determines actuator torque rating and battery capacity
Gait modes	Static walk, trot, turn-in-place	Determines controller update rate and sensor requirement

3. Mechanical Subsystem

3.1 Torso / Chassis

A rectangular parallelepiped torso with corners removed for weight reduction was selected as the chassis concept, following the structural approach documented for 3D-printed STM32-based quadruped prototypes ^[4]. A reference torso of approximately 210 mm length, 96 mm width, and 60–70 mm height was scaled proportionally to fit the 50–70 cm overall length target ^[4]. All four hip actuators were mounted flush against the torso rather than at the knee, following the general design rule established in quadruped scaling studies that reflected leg inertia is minimized when actuator mass is kept close to the body ^[5].

3.2 Leg Design

Each leg was assigned 3 DOF (hip abduction/adduction, hip flexion/extension, and knee flexion), for a total of 12 DOF across four legs; two additional DOF were allocated to a spine/waist yaw joint and a head pan joint to reach the 14 DOF upper target ^[6]. Thigh and calf link lengths were set at a ratio consistent with published quadruped leg designs (approximately 50 mm : 80 mm at small scale), scaled up to match the 30–40 cm standing height requirement ^[4]. Hip joint range of motion was set to -90° to +90°, and knee range of motion was set to approximately 0° to

150°, matching the joint range constraints used in comparable canine-inspired quadruped designs ^[7]. Motors were placed proximally at the hip with the femur driven through a rigid linkage, a configuration adopted from the Emily quadruped and OpenDoggo builds to reduce swing-phase inertia ^{[8][9]}.

3.3 Foot Design

A compliant rubber or silicone foot pad with a passive elastomer damper was specified at each foot tip to absorb touchdown impact and improve ground contact stability, following the mechanical design and simulation recommendations reported for quadruped robot motion control ^[10].

3.4 Structural Materials

- Torso and leg links: PLA/PETG 3D-printed structure, consistent with low-cost quadruped prototypes reported in literature ^{[4][7]}
- Load-bearing hip brackets: aluminum or reinforced nylon for fatigue resistance under repeated gait cycles
- Fasteners: M3 threaded inserts recommended for printed parts to avoid thread stripping under vibration

4. Actuation Subsystem (Motors)

Joint torque requirements were estimated using the standing and trotting torque models reported for quadruped robots, in which the static hip/knee torque scales with robot mass, leg length, and the number of legs in ground contact, and dynamic trot torque was calculated using $\tau = \frac{1}{4} \cdot g \cdot m_{\text{robot}} \cdot l_{\text{ls}} \cdot \sin(\theta_2(t)/2) \cdot (h_{\text{max}}/l_{\text{ls}} + 1)$ ^[11]. Applying this model to a 14 kg mass budget indicated that hobby-class 9 g or MG90S servos, as used in lightweight sub-2 kg builds, were insufficient, and that mid-torque smart servos or brushless geared actuators with integrated encoders were required ^[12].

4.1 Recommended Actuator

- Primary joint actuators: Dynamixel MX-series or XM430-series smart servos, selected for their built-in position/velocity/current control, magnetic encoder feedback, and daisy-chain TTL/RS-485 bus, consistent with mechanical design studies that specified ROBOTIS Dynamixel MX-series actuators for a comparable quadruped class ^[13]
- Alternative lower-cost actuator: high-torque digital servo with metal gears and dual bearings (>20 kg·cm stall torque), following servo-selection guidance for quadruped builds that recommended digital metal-gear servos over analog/plastic-gear types for load-bearing joints ^[2]
- Quantity: 12 actuators for the primary leg DOF, plus 2 additional actuators for spine/head DOF where 14 DOF is implemented

4.2 Torque Sizing Result

Per-leg static torque was calculated by dividing half of the total robot weight across the two legs in ground contact during trot swing phase, following the leg torque tutorial method for legged robots [14]. This calculation confirmed that hip and knee actuators required a minimum continuous torque rating in the range used by MX-series Dynamixel actuators, rather than the lower torque ratings adequate only for sub-2 kg robots ^{[11][13]}.

5. Electrical Subsystem

5.1 Main Controller

A two-tier controller architecture was selected: a real-time microcontroller was assigned to handle joint PWM/servo-bus timing and gait scheduling, while a single-board computer was assigned to handle higher-level planning, and this split follows the documented advantage of offloading real-time actuation tasks to a dedicated MCU while a companion computer handles perception and AI ^[15].

Component	Recommended Part	Function
Low-level MCU	STM32F4-series (or ESP32 for lower-cost builds)	Servo bus control, gait timing, sensor polling, real-time joint control [15]
High-level SBC	Raspberry Pi 4/5	Inverse kinematics, gait planning, camera vision, Wi-Fi/remote command interface [15][16]
Servo/Motor driver bus	Dynamixel TTL/RS-485 bus or PCA9685 PWM driver board	Simultaneous multi-servo control from a single MCU bus, as implemented in a 12-DOF ESP32-based quadruped [17]

5.2 Power Subsystem

- Main battery: 3S or 4S Lithium-Polymer pack (11.1V–14.8V), sized to the current draw of 12–14 smart servos under load
- Power distribution board with buck converters: 5V rail for logic/SBC, and a separate high-current rail for servo power, to isolate motor electrical noise from the control electronics
- Battery monitoring: voltage/current sensor on the main power rail for low-battery cutoff and telemetry

5.3 Wiring and Communication

- Joint actuator bus: daisy-chained TTL half-duplex bus (Dynamixel protocol) or individual PWM lines through a PCA9685 I2C driver, mirroring the wiring approach reported for a 12-DOF servo-based quadruped ^[17]

- MCU-to-SBC link: UART or SPI for low-latency command exchange between the real-time controller and the companion computer ^[15]
- External interface: Wi-Fi (onboard SBC) and optional Bluetooth for remote monitoring/control

6. Sensor Subsystem

The sensor suite was specified to provide the three data types required for closed-loop quadruped locomotion: body orientation, joint state, and ground contact/force feedback, consistent with the sensor suites documented across multiple quadruped robot platforms ^[18].

Sensor	Recommended Part	Purpose
IMU (9-axis)	MPU-9250 or BNO055	Body roll/pitch/yaw and angular velocity for balance control, following the IMU placement used in quadruped foot/body sensing research [19][18]
Joint encoders	Integrated magnetic encoders in smart servo (Dynamixel) or external rotary encoder on geared DC motor	Real-time joint angle and velocity feedback for closed-loop position control, as used across all reviewed quadruped platforms [18][20]
Foot contact/force sensor	Force-sensitive resistor (FSR) at each foot tip	Ground contact detection and reaction force estimation to trigger stance/swing phase transitions in the gait state machine, following FSR-based contact sensing proposed for quadruped locomotion policies [21]
Ultrasonic/ToF distance sensor	HC-SR04 or VL53L0X	Obstacle detection for basic autonomous navigation, following the sonar sensor used in a comparable 12-DOF ESP32 quadruped build [17]
Camera (optional)	ESP32-CAM or Raspberry Pi Camera Module	Vision-based perception for object/obstacle recognition on the SBC tier [17]

6.1 Sensor Fusion

Foot contact state was determined using a threshold function applied to the FSR/ground-reaction-force reading, in which contact was declared when the measured force exceeded a calibrated threshold value, following the contact-detection formulation reported for legged

robot terrain sensing ^[21]. Body orientation was estimated by fusing IMU accelerometer and gyroscope data through a complementary or Kalman filter, consistent with the proprioceptive state-estimation pipelines documented for quadruped odometry ^[22].

References

1. Emily, a Quadruped Dog Robot: 12 Steps — Instructables project documentation.
2. Servo motor considerations for a quadruped — RobotShop community discussion on digital vs. analog servo selection for quadruped legs.
3. Open Doggo — 12-DOF open-source hobby-servo quadruped robot dog, HackMD documentation.
4. Complexity 9663746 — STM32-based, 3D-printed quadruped robot structural design paper.
5. Design and Scaling of Versatile Quadruped Robots — Robotica journal (2024) actuator placement and inertia scaling study.
6. Mechanical Design, Analysis and Simulation of a Quadruped Robot — design constraints and 12-DOF leg configuration reference.
7. Design of a Canine Inspired Quadruped Robot (DoggyDeux) as a Platform — joint range of motion and 3D-printed frame design.
8. Emily, a Quadruped Dog Robot — proximal actuator mounting and femur linkage design.
9. Open Doggo — servo placement at torso level to reduce leg inertia.
10. Mechanical Design and Simulation Studies of a Quadruped Robot Motion Control — Euler-Lagrange joint torque and foot design analysis.
11. Design and Scaling of Versatile Quadruped Robots — trot gait torque model (Robotica, 2024).
12. Design, Analysis and Fabrication of Quadruped Robot with 3-DOF Legs (IJISSET) — joint torque calculation for standing and trotting.
13. Mechanical Design, Analysis and Simulation of a Quadruped Robot — Dynamixel MX-series actuator specification.
14. Robot Leg Torque Tutorial — RobotShop Learning Center, leg torque calculation methodology.
15. Comparison of MCUs — Raspberry Pi + STM32 architecture discussion for real-time robot control.
16. ROS 2 Robot Kit Buying Guide — Raspberry Pi vs. Jetson selection criteria for robot companion computers.
17. Quadruped robot dog with 12DOF and IK — ESP32-WROOM-CAM based 12-DOF servo quadruped with PCA9685 driver and HC-SR04 sonar.
18. Complete Quadruped Robots Guide 2026 — IMU, joint encoder, and foot force sensor suite reference.
19. Whole-body kinematic and dynamic modeling for quadruped robot under different gaits — foot force sensor and IMU placement.

20. Enhanced State Estimation for quadruped robots — IMU, joint encoder, and foot contact sensor fusion.
21. Ground contact and reaction force sensing for linear policy quadruped robots — FSR-based contact threshold formulation.
22. Odometry and Contact Estimation for Legged Robots — proprioceptive ESEKF state estimation pipeline.
23. A Leg Design Method for High Speed Quadrupedal Locomotion — inverse kinematics and leg trajectory derivation.
24. Probabilistic Contact Estimation for Legged Robots — contact-based finite state machine for gait phase transition.